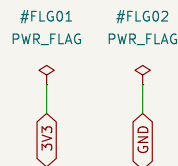
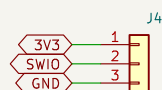
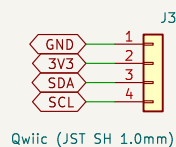
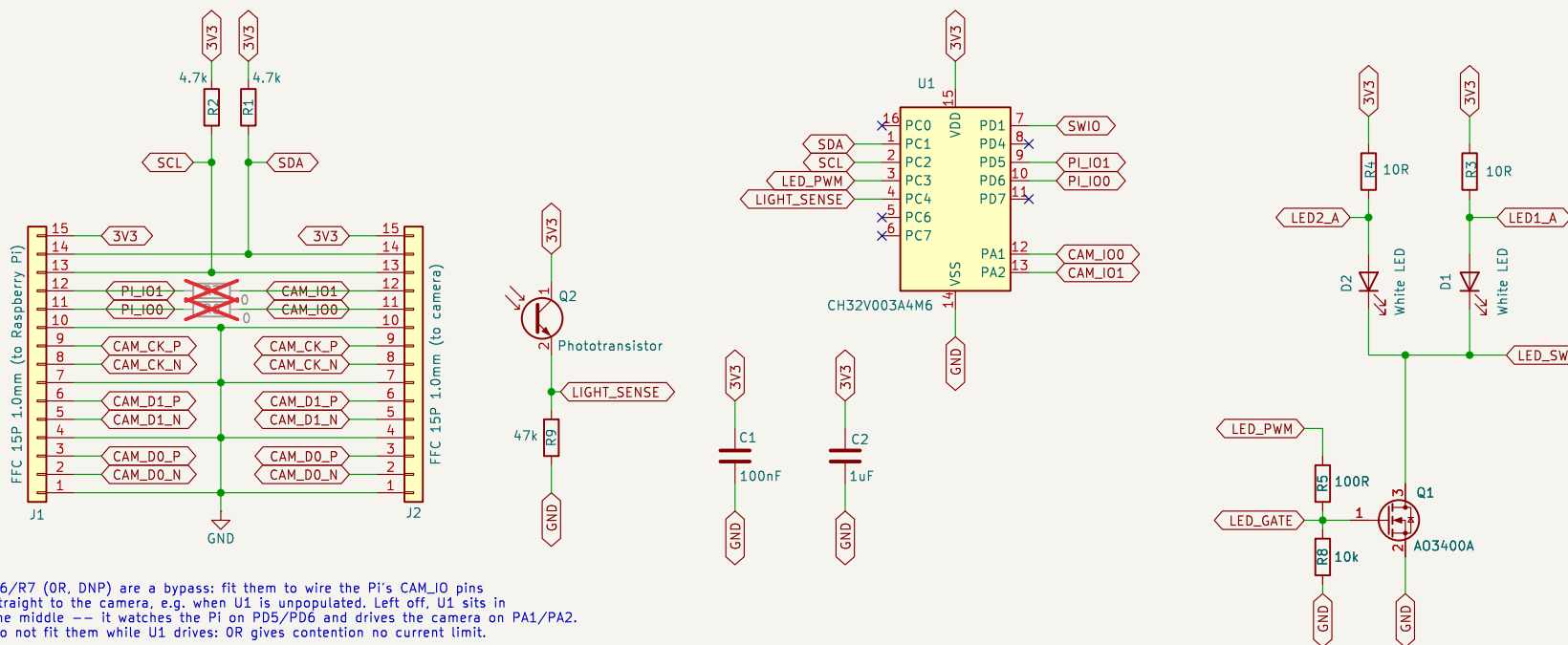


RPi Camera LED interposer: sits between a Raspberry Pi and its camera.  
 MIPI lanes, I2C, GND and 3V3 pass straight through J1 -> J2.  
 U1 (I2C slave on the camera bus) drives the camera GPIO pins 11/12,  
 PWM-drives the illumination LEDs and reads ambient light on its ADC.

LED current is set by R3/R4 against the 3V3 rail (about 30-40mA each with  $V_f=2.9V$ ). Total draw comes from the Pi camera 3V3 supply - keep the duty/current within the Pi's camera connector budget.



All parts chosen for JLCPCB / NextPCB Rev0 assembly; see research/\*.md for availability and pricing (checked 2026-09-01).

SOP-16 bonds no two ports to one pin, so PD1/SWIO is programming only and PC3/T1CH3 drives the LED gate; no LED flicker while flashing firmware. PC6, PC7, PD4 and PC0 are spare. PD7 is left unused on purpose: it is NRST until the RST\_MODE option byte says otherwise, so a Pi GPIO on it would reset the MCU on a factory-fresh part.

CH32V003A4M6 camera GPIO / illumination / light sensor

Sheet: /  
 File: rpi-camera-led.kicad\_sch

**Title: RPi Camera LED interposer**

Size: A4 Date: 2026-09-01

KiCad E.D.A. 9.0.7

Rev: A

Id: 1/1